

Bangladesh University of Business and Technology

Department of Computer Science and Engineering

LAB REPORT

Course Name: Computer Graphics Lab

Course Code: CSE 342

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Intake: 51, Section: 03

Submitted To:

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Experiment 01: Implementation of DDA Line Drawing Algorithm

Objective:

1. Understand the working principle of DDA.
2. Generate intermediate pixels between two coordinates.

Mathematical Formulation:

Slope $m = dy / dx$

If $|m| \leq 1$: $x_{next} = x + 1$, $y_{next} = y + m$

If $|m| > 1$: $y_{next} = y + 1$, $x_{next} = x + (1/m)$

Experiment 02: Implementation of Bresenham's Line Drawing Algorithm

Objective:

To generate intermediate pixel positions efficiently using integer calculations.

Decision Parameter Logic:

Initial $p = 2dy - dx$

If $p < 0$: $p_{next} = p + 2dy$

Else: $p_{next} = p + 2(dy - dx)$, $y_{next} = y + 1$

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Experiment 04: Basic 2D Transformations

Theory:

Transformations like Translation, Scaling, Rotation, and Shearing are represented using matrix operations.

Transformation Matrices:

Translation: $\begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix}$

Scaling: $\begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Rotation: $\begin{bmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Shearing: $\begin{bmatrix} 1 & Sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$